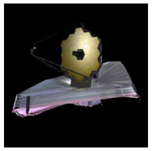

8663 - Is the Moon an Active Source of Near-Earth Asteroids? Testing the Origins of Earth Quasi-Satellite Kamo`oalewa

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	2		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	3		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	4		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	5		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	6		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	7		NIRSpec IFU Spectroscopy	(1) Kamooalewa
	8		NIRSpec IFU Spectroscopy	(1) Kamooalewa

ABSTRACT

We propose NIRSpec IFU observations of (469219) Kamoʻoalewa, a unique Earth quasi-satellite asteroid discovered in 2016. This is the first quasi-satellite to have its physical properties studied to assess its composition and origins. Analysis of previous observations with 8m-class ground observatories produced the hypothesis that it may represent the first known asteroid to have formed from lunar collisional ejecta. However, Kamoʻoalewa's faintness limits any stricter tests of its material composition from ground-based observatories. Such tests require knowledge of its silicate mineralogy, the presence or absence of organic or hydrated material(s), and measurements of its size/albedo. By observing Kamoʻoalewa with NIRSpec, we will comprehensively characterize the presence of these materials and provide the first measurements of its size and albedo. This program will enable strict tests of the possible lunar origin for this object, testing whether the moon acts as an entirely unpredicted source of near-Earth asteroids in conflict with fundamental population models.

OBSERVING DESCRIPTION

The proposed observations target near-Earth asteroid (469219) Kamo`oalewa. We propose to use NIRSpec in its IFU Prism mode to characterize this Earth quasi-satellite asteroid, which varies in brightness by about 1.5 magnitudes as its viewing geometry changes over the course of annual observing windows. To ensure an efficient use of telescope time, we restrict our observing window to the two months where the quasi-satellite is brighter than V=23. We note that despite this object's close proximity to Earth, and its overall faintness, its orbit is well characterized (with plane of sky 3-sigma uncertainties < 0.2 square arcseconds) and is well within JWST's slew rates (~ 30 milliarcseconds/sec). The groups, integrations, and nod patterns are selected to ensure SNR > 33 near 3.0 microns.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.3454527106666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Observation

Proposal 8663, Observation 1

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 1) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 2

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 2) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 3

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 3) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 4

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 4) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 5

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 5) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 6

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 6) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 7

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 7) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075

Observation

Proposal 8663, Observation 8

Wed Dec 03 00:00:12 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Diagnostics

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Observation 8) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

Solar System Targets

#	Name	Level 1	Level 2	Level 3
(1)	Kamooalewa	TYPE=ASTEROID,A=1.00102447694204,E=0.1040534310625292,I=7.773894173631178,O=66.43991583004482,W=307.0951007739783,M=187.6486415815915,EQUINOX=J2000,EPOCH=11-APR-2017:00:00:00,EpochTimeScale=TDB,R0=1.,A1=0.,A2=-1.345452710666E-13,A3=0.,ALN=1.,NM=2.,NN=5.093,NK=0.,AMRAT=0.		

Comments: Extended=Unknown

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	NRSIRS2RAPID	44	1	false	true	NONE	4	4	2626.0	221864

Special Requirements

Between Dates 29-JAN-2026:00:00:00 and 13-APR-2026:00:00:00

DEFAULT WINDOW: ANGULAR RATE Kamooalewa FROM JWST LESS THAN 0.075